

Conversion Log

Production Conversion Log

January 2025 · Girardot, Colombia

Submission reference	RTF0-310825
Reporting period	January 2025 (2025-01)
Plant	Girardot, Colombia
Product	DEV-P100 — refined pyrolysis oil
Off-taker	Crown Oil Limited (UK)
Regulator	UK DfT — LCF Delivery Unit
Density — EU-grade	0,756 kg/L
Density — PLUS-grade	0,856 kg/L
Production days observed	21 of 31 (10 non-production days)
Source	daily_production table (DFT backend, snapshot 2026-05-21T00:00:00Z)
Generated	2026-05-21T00:00:00Z

Signed for and on behalf of OisteBio GmbH

Paolo Ughetti

Managing Director

Date: _____

Executive Summary

Per-day kilogram-to-litre conversion log for **January 2025** production at the Girardot, Colombia pyrolysis facility, operated by **OisteBio GmbH**. Output product is **DEV-P100 — refined pyrolysis oil**, supplied to **Crown Oil Limited (UK)** under the UK DfT — LCF Delivery Unit reporting framework. Litres are derived from kilograms via the nominal product densities listed below, as materialised by the `litres_eu` and `litres_plus` GENERATED ALWAYS columns on the `daily_production` table (Alembic migration `0007_persist_production_litres.py`).

Monthly aggregate

Metric	Value
Feedstock routed to thermal conversion (input)	3 043 312,000 kg
Total EU-grade production	760 828,000 kg / 1 006 386,241 L
Total PLUS-grade production	1 308 624,160 kg / 1 528 766,539 L
Liquid-product yield (EU + PLUS) over input	68,00 %
Grand total volume (EU + PLUS)	2 535 152,780 L

Co-products tracked separately on the same daily ledger (not converted to litres because they are solid / gaseous): carbon black **426 063,680 kg**, steel scrap **286 071,328 kg**, water condensate **51 736,304 kg**, syngas **144 557,318 kg**, recorded losses **65 431,203 kg**. Mass-balance closure across all streams is reported in Annex A.

Daily Conversion — January 2025

One row per production day from the `daily_production` table. Days with no production record (weekends, public holidays, maintenance) are listed in the gap line below. Litres computed at constant density: **EU 0,756 kg/L, PLUS 0,856 kg/L**. Liquid yield = (EU kg + PLUS kg) / input kg.

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-01-04	121 242,000	30 310,500	52 134,060	0,756	0,856	40 093,254	60 904,276	68,00 %	—
2025-01-06	137 180,000	34 295,000	58 987,400	0,756	0,856	45 363,757	68 910,514	68,00 %	—
2025-01-07	150 632,000	37 658,000	64 771,760	0,756	0,856	49 812,169	75 667,944	68,00 %	—
2025-01-08	140 093,000	35 023,250	60 239,990	0,756	0,856	46 327,050	70 373,820	68,00 %	—
Total	3 043 312,000	760 828,000	1 308 624,160	—	—	1 006 386,241	1 528 766,539	68,00 %	

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-01-09	145 235,000	36 308,750	62 451,050	0,756	0,856	48 027,447	72 956,834	68,00 %	—
2025-01-10	146 915,000	36 728,750	63 173,450	0,756	0,856	48 583,003	73 800,759	68,00 %	—
2025-01-13	158 292,000	39 573,000	68 065,560	0,756	0,856	52 345,238	79 515,841	68,00 %	—
2025-01-14	147 889,000	36 972,250	63 592,270	0,756	0,856	48 905,093	74 290,035	68,00 %	—
2025-01-15	141 214,000	35 303,500	60 722,020	0,756	0,856	46 697,751	70 936,939	68,00 %	—
2025-01-16	148 245,000	37 061,250	63 745,350	0,756	0,856	49 022,817	74 468,867	68,00 %	—
2025-01-17	136 842,000	34 210,500	58 842,060	0,756	0,856	45 251,984	68 740,724	68,00 %	—
2025-01-20	153 240,000	38 310,000	65 893,200	0,756	0,856	50 674,603	76 978,037	68,00 %	—
2025-01-21	152 708,000	38 177,000	65 664,440	0,756	0,856	50 498,677	76 710,794	68,00 %	—
2025-01-22	141 993,000	35 498,250	61 056,990	0,756	0,856	46 955,357	71 328,259	68,00 %	—
2025-01-23	151 976,000	37 994,000	65 349,680	0,756	0,856	50 256,614	76 343,084	68,00 %	—
2025-01-24	140 403,000	35 100,750	60 373,290	0,756	0,856	46 429,563	70 529,544	68,00 %	—
2025-01-27	146 115,000	36 528,750	62 829,450	0,756	0,856	48 318,452	73 398,890	68,00 %	—
2025-01-28	160 323,000	40 080,750	68 938,890	0,756	0,856	53 016,865	80 536,086	68,00 %	—
2025-01-29	142 639,000	35 659,750	61 334,770	0,756	0,856	47 168,981	71 652,769	68,00 %	—
2025-01-30	153 252,000	38 313,000	65 898,360	0,756	0,856	50 678,571	76 984,065	68,00 %	—
2025-01-31	126 884,000	31 721,000	54 560,120	0,756	0,856	41 958,995	63 738,458	68,00 %	—
Total	3 043 312,000	760 828,000	1 308 624,160	—	—	1 006 386,241	1 528 766,539	68,00 %	

Non-production days (10): 2025-01-01, 2025-01-02, 2025-01-03, 2025-01-05, 2025-01-11, 2025-01-12, 2025-01-18, 2025-01-19, 2025-01-25, 2025-01-26. These dates have no row in daily_production and are reported as N/A by omission. Weekends and public holidays in Colombia during January 2025 account for all of them.

Methodology & References

Volumetric conversion applies the canonical relation $\text{litres} = \text{kg} / \text{density}$ at the calibrated product density measured at 15 °C for the reporting month. Densities are resolved per production month from the platform product_densities table (one row per

(product, effective_from) pair), so historical conversions remain reproducible if downstream batches recalibrate the product specification. The densities applied to January 2025 are:

- **EU-grade (DEV-P100-EU)** — $\rho_{\text{EU}} = 0,756 \text{ kg/L}$
- **PLUS-grade (DEV-P100-PLUS)** — $\rho_{\text{PLUS}} = 0,856 \text{ kg/L}$

Both density values are ISCC-recognised product-specification figures for the OisteBio DEV-P100 — refined pyrolysis oil family. They are seeded and maintained in the platform product_densities table — initially loaded by Alembic migration 0005_seed_product_densities.py and extended with per-month effective rows by 0014_monthly_densities.py. Once a reporting period closes, the row applicable to that month becomes immutable for audit purposes; new effective rows are appended forward, never back-dated.

Per-row litre values shown in the daily table are not stored independently — they are derived deterministically inside PostgreSQL as **GENERATED ALWAYS** columns on the daily_production table:

```
litres_eu    NUMERIC GENERATED ALWAYS AS (eu_prod_kg / 0.756) STORED,
litres_plus  NUMERIC GENERATED ALWAYS AS (plus_prod_kg / 0.856) STORED
```

These columns are introduced by Alembic migration 0007_persist_production_litres.py. Once persisted, historical litre values become immutable for audit purposes even if product densities are revised in the future. Write access to litres_eu and litres_plus is prohibited at the schema level: any attempt to update them raises PostgreSQL error code 42601.

The **input kg** column reports the daily value of daily_production.kg_to_production — the mass of pre-conditioned ELT feedstock routed to the pyrolysis reactor that calendar day. The **yield %** column reports the fraction of input mass that exits the refining line as liquid product (EU + PLUS), with the remainder distributed across carbon black, steel scrap, water condensate, syngas, and recorded losses (totals reported in the Executive Summary above and reconciled in Annex A).

This Production Conversion Log accompanies Annex A (mass balance) and Annex D (closing-stock carry-over) in the RTFO-310825 submission bundle and is cryptographically anchored to the cover letter via the SHA-256 digest reported in the running footer (short prefix 000000000000). The authoritative digest is reproduced in MANIFEST.sha256 at the bundle root.

— End of Production Conversion Log —